

Module 03: Perception in High School

Learning Objectives

1. Define **Perception** within the context of High School.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In High School, we see Perception manifest in: * **Specific Example 1:** Bayesian updating in statistics is the mathematical foundation of Active Inference perception: when you flip a coin 10 times and get 7 heads, your prior belief ($P(\text{fair}) = 0.5$) combines with the likelihood of the observed data to produce a posterior distribution that shifts toward "biased coin" -- this is exactly how a perceptual system updates its generative model given surprising sensory evidence. * **Specific Example 2:** In signal processing, a Kalman filter estimates the true position of a moving object from noisy GPS measurements by maintaining a predicted state (generative model) and weighting it against each new observation based on measurement precision -- this is perception as mathematical inference, and the filter's update equation is a concrete implementation of prediction error minimization.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.